

LD9 Business Interpretation Summary

Final Documentation Package for the Bank Marketing Subscription Prediction Project

Prepared for: Ruturaj Mokashi

This document explains the project results from a business point of view. It focuses on what the models mean for a marketing team that wants to predict whether a bank customer will subscribe to a term deposit.

1. Business Context

The business problem is a classification task: predict whether a customer will subscribe to a term deposit after a marketing contact. A strong model helps the business focus outreach on customers who are more likely to subscribe, reducing wasted effort and improving campaign efficiency.

2. What Each Notebook Added to the Business Story

Notebook	Business meaning
Notebook 1	Source audit, coverage matrix, dataset selection, project roadmap, and theory mapping.
Notebook 2	Data preparation, train/test split, preprocessing pipeline, target balance check, and baseline models.
Notebook 3	Decision Tree vs Random Forest, bagging logic, OOB score, Random Subspace Method, feature importance, and tree-count stability.
Notebook 4	Cross-validation, StackingClassifier with LR and RF meta-models, and model diversity analysis.
Notebook 5	Manual OOF stacking, blending, and comparison between OOF stacking and blending.
Notebook 6	Three-layer stacking, final complexity-vs-value decision, and Gradient Boosting.
Notebook 7	Final MVD proof checklist, final model comparison, and final business recommendation.

3. Key Final Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
StackingClassifier (RF meta)	0.8598	0.8176	0.9064	0.8597	0.9247
Three-Layer Stacking	0.8612	0.8369	0.8781	0.8570	0.9251
Manual OOF Stacking	0.8603	0.8354	0.8781	0.8562	0.9259

StackingClassifier (LR meta)	0.8589	0.8338	0.8771	0.8549	0.9265
Blending 70/30	0.8554	0.8314	0.8715	0.8509	0.9232
Gradient Boosting	0.8325	0.8109	0.8431	0.8267	0.9123

The strongest practical model was the StackingClassifier with a Random Forest meta-model. It gave the best test F1-score (0.8597) and the highest recall (0.9064). This means it was especially good at catching likely subscribers while still keeping overall performance strong.

4. Final Business Recommendation

- Use the StackingClassifier with Random Forest meta-model as the main recommended model.
- Use F1-score as the main comparison metric because it balances precision and recall.
- Use recall as a secondary business metric because missing true subscribers can reduce campaign value.
- Keep probability-based ranking (ROC-AUC) in mind when the business wants to prioritize customers by likelihood score rather than only yes/no classification.

5. Operational Interpretation of the Best Model

- High recall means the model finds most of the real subscribers.
- Reasonable precision means the model does not over-target too many unlikely customers.
- A strong F1-score means the model balances both concerns well.
- The stacker combines the strengths of different models instead of depending on a single model type.

6. Why We Did Not Choose the More Complex Three-Layer Stack

The three-layer stack achieved the highest test accuracy (0.8612), but its F1-score was slightly lower than the simpler RF-meta stacker. In business work, more complexity is only useful if it gives a clear gain. Here, the extra complexity did not provide enough additional value.

7. Suggested Next Business Steps

1. Deploy the recommended stacker in a scoring workflow for new campaign lists.
2. Set a probability threshold based on campaign budget and capacity.
3. Monitor campaign conversion rates, precision, and recall after deployment.
4. Retrain the model periodically as campaign behavior changes.